

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Soils: fertility, erosion & management - Set 1

Mark scheme

AQA spec 3.2.5; The physical environment; textbook pp. 158-169

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark each 1 mark for each of the 3 points below. 1 B (about 40:40:20 sand:silt:clay) 1 Balanced mix gives good drainage and aeration from the sand fraction 1 Water retention and nutrient (ion) retention from the clay fraction – easy root penetration and cultivation Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • loam → a balanced mixture of sand, silt and clay	3

Question	Marking guidance	Marks
01.2	1 mark each 1 mark for each of the 3 points below. 1 Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) 1 Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand 1 With only 5% clay there is little surface to hold ions so nutrients are leached – low water retention too Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. permeability → how easily water passes through / lets water through adsorb → nutrient ions sticking to the surfaces of clay and humus leached → nutrients or ions washed down out of the soil / washed away by water	3
01.3	1 mark each 1 mark for each of the 2 points below. 1 C (clay) holds much water so has a high thermal capacity: warms and cools slowly – A warms rapidly after cold weather 1 Clay particles held together by capillary water so roots find it hard to push between them – penetration is easy in sandy A Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round.	2
01.4	1 mark each 1 mark for each of the 2 points below. 1 Shake a known mass of soil with water in a measuring cylinder and leave to settle – particles settle in order of size (sand first, then silt, then clay; organic matter floats) 1 Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	

Question	Marking guidance	Marks
02.1	Marks shared Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. up to 2 Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients – worms create tunnels improving drainage and aeration up to 2 Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds up to 2 Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions – nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption up to 2 Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • ammonification → ammonium released from dead organic matter and protein • humus → dark, decomposed organic matter in soil • peds → soil crumbs / aggregates of soil particles	6
02.2	1 mark each 1 mark for each of the 2 points below. 1 This is the range of tolerance of most plants and soil biota – acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes 1 In alkaline soils phosphates become insoluble / unavailable Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • range of tolerance → the range of conditions in which a species can survive	2
02.3	1 mark each 1 mark for each of the 2 points below. 1 Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate 1 Infiltration falls so waterlogging is more likely and runoff / erosion increase Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Interstitial → pore spaces / air spaces between the particles • nitrification → ammonium oxidised to nitrite and then nitrate • Infiltration → water soaking into the soil surface	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.	

Question	Marking guidance	Marks
03.1	1 mark each 1 mark for each of the 2 points below. 1 Soil loss increases as vegetation cover decreases: forest lowest (0.5), bare soil highest ($60 \text{ t ha}^{-1} \text{ yr}^{-1}$) - over 100 times more 1 Ploughing downslope loses nearly three times as much as contour ploughing for the same crop Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. contour ploughing → ploughing along the slope rather than up and down it	2
03.2	Any 4 of 5 1 mark each for any 4 of the 5 points below, in any order, maximum 4. 1 Leaves / leaf litter intercept raindrops so soil particles are not dislodged by rain splash 1 Roots bind soil particles together 1 Vegetation acts as a windbreak reducing wind velocity and the kinetic energy available to carry particles 1 Plants increase infiltration (root channels, organic matter) so less surface runoff 1 Organic matter / humus from vegetation binds particles into stable peds Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. windbreak → a line of trees or hedge that slows the wind infiltration → water soaking into the soil surface humus → dark, decomposed organic matter in soil Do not credit - Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
03.3	1 mark each 1 mark for each of the 2 points below. 1 Furrows running downslope act as channels / gulleys that increase the velocity (kinetic energy) of runoff 1 Furrows along the contour trap the water so it loses energy, deposits particles and infiltrates Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. contour ploughing → ploughing along the slope rather than up and down it	2
03.4	1 mark each 1 mark for each of the 2 points below. 1 In the dry season the soil (especially low-clay soils) is loose and dry with little cohesion so wind blows it away 1 In the wet season heavy rain causes rain splash and surface runoff when infiltration capacity is exceeded Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. infiltration → water soaking into the soil surface	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
04.1	2 marks per point 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. 2 Overgrazing: vegetation is eaten faster than it grows so soil is exposed – hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff 2 Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles – fewer worms so less aeration and lower infiltration capacity – fewer nutrients released so less vegetation cover Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • infiltration → water soaking into the soil surface • humus → dark, decomposed organic matter in soil Do not credit • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
04.2	1 mark each 1 mark for each of the 2 points below. 1 Tree roots that held the soil together decompose so the soil is bound less strongly 1 Heavy rain adds mass and lubricates particles – without interception more water enters the soil, making slumping more likely Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2

Question	Marking guidance	Marks
04.3	Any 4 of 6 1 mark each for any 4 of the 6 points below, in any order, maximum 4. 1 Sedimentation in rivers reduces flow capacity so flooding downstream is more likely 1 Sedimentation in reservoirs reduces their storage volume 1 Coastal sedimentation covering reefs / seagrass – turbidity reducing light for aquatic plants 1 Increased atmospheric particulates from wind erosion worsening asthma – dust covering crops / urban areas 1 Desertification as vegetation cannot re-establish – reduced rainfall 1 Nutrients (phosphates) carried on soil particles causing eutrophication Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • particulates → fine solid particles / smoke and dust • eutrophication → nutrient enrichment of water leading to algal growth Do not credit • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
05.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 $110 \times 0.67 \times 1.35 \times (0.50 \times 0.90) \times 1.0$ 1 = $44.8 \text{ t ha}^{-1} \text{ yr}^{-1}$ (accept 44.7-44.8) Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit ($\text{t ha}^{-1} \text{ yr}^{-1}$) is already printed on the answer line. Do not credit • A bare wrong answer with no working.	2

Question	Marking guidance	Marks
05.2	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 $110 \times 0.67 \times 1.35 \times (0.35 \times 0.60) \times 0.25$ 1 $= 5.2 \text{ t ha}^{-1} \text{ yr}^{-1}$ Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit ($\text{t ha}^{-1} \text{ yr}^{-1}$) is already printed on the answer line. • contour ploughing → ploughing along the slope rather than up and down it Do not credit • A bare wrong answer with no working.	2
05.3	1 mark each 1 mark for each of the 3 points below. 1 Beans / up-and-down ploughing: severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall 1 Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable 1 Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land ($C = 0.02$) needed to bring loss below $1 \text{ t ha}^{-1} \text{ yr}^{-1}$ Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • sustainability → meeting present needs without stopping future generations meeting theirs • terracing → cutting the slope into level steps Do not credit • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
05.4	1 mark each 1 mark for each of the 2 points below. 1 Factors are averages / arbitrary values for typical conditions – rainfall erosivity varies between years / storms 1 Does not include gully erosion, landslides, wind erosion or local variation in soil – based on data from particular regions Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.5	1 mark for the point below. 1 C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.	

Question	Marking guidance	Marks
06.1	1 mark each 1 mark for each of the 5 points below. 1 Terracing: sloping land divided into narrow level fields held by retaining walls along contours – water slows as it crosses each terrace and infiltrates instead of running off 1 Tied ridging: on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff 1 Windbreaks: hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles – protect bare fields after harvest 1 Mulching: organic matter on the surface protects soil from raindrop impact and wind – decomposes to humus which binds particles – retains moisture 1 Strip cropping: strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field – slows runoff Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Terracing → cutting the slope into level steps • infiltration → water soaking into the soil surface • humus → dark, decomposed organic matter in soil Do not credit • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	5
06.2	1 mark each 1 mark for each of the 2 points below. 1 No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles 1 Permanent root systems and canopy cover protect the soil all year Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	Any 3 of 4 1 mark each for any 3 of the 4 points below, in any order, maximum 3. 1 Control stocking density so vegetation regrows faster than it is eaten 1 Restrict access to vulnerable areas such as river banks (fencing) 1 Move / remove livestock at high-risk times, eg after heavy rain when soil is wet 1 Rotational grazing to rest pastures Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.	

Question	Marking guidance	Marks
07.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 Water = $65.0 - 55.0 = 10.0$ g – fresh soil = 40.0 g 1 $10.0 / 40.0 \times 100 = 25.0\%$ Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. Do not credit • A bare wrong answer with no working.	2
07.2	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 Organic matter = $55.0 - 52.6 = 2.4$ g – dry soil = 30.0 g 1 $2.4 / 30.0 \times 100 = 8.0\%$ Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. Do not credit • A bare wrong answer with no working.	2
07.3	1 mark each 1 mark for each of the 2 points below. 1 So all water is removed and the mass loss on burning is due to organic matter only 1 Heating until mass no longer changes shows all water has evaporated Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	1 mark each 1 mark for each of the 3 points below. 1 Organic matter releases nutrients as it decomposes and provides food for soil biota 1 Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration 1 Surface organic matter / humus adhesion reduce rain splash and wind erosion Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • humus → dark, decomposed organic matter in soil • peds → soil crumbs / aggregates of soil particles	3

Question	Marking guidance	Marks
07.5	1 mark for the point below. 1 Samples taken from different depths / locations (organic matter is concentrated near the surface) – incomplete burning – different heating times Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	
08.1	Correct answer only 1 mark for the correct option. No mark if more than one box is ticked. 1 B - tiny pores between clay particles allow capillary rise – none in sand	1
08.2	1 mark each 1 mark for each of the 3 points below. 1 Hygroscopic: held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants 1 Capillary: held in pores by surface tension – does not drain but can be absorbed by plants 1 Gravitational: drains freely through large pores – available to plants only briefly as it passes Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.3	1 mark for the point below. 1 Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • peds → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	